

RT_GEOMETRIC_CARRIER_MODEL_FOR_P0MU_R_v1

Title: Geometric carrier model behind RT's P0mu_R closure-gate coefficient

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Project: Rhythm Theory (RT)

Status: public companion to CLAIM_ALPHA_CLOSURE_GATE_v39

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Scope: geometry/readout carrier for the v39 ledger; not a full metric 3D physics derivation

Canonical public PDF after publication:

https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/RT_GEOMETRIC_CARRIER_MODEL_FOR_P0MU_R_v1.pdf

Primary claim document:

https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/CLAIM_ALPHA_CLOSURE_GATE_v39.pdf

Repository / Release branch:

<https://github.com/Bo-Fremling/rhythm-theory-release/tree/Release>

0. Purpose

This file supplies the public geometric carrier picture that is only summarized in CLAIM_ALPHA_CLOSURE_GATE_v39.

It explains why the v39 words:

surface, seam, port, boundary, current, retained, activated seam, C90, G39

are not just decorative labels. They refer to a typed discrete readout carrier.

However, this file also keeps the claim honest:

This is a discrete carrier/readout geometry.
It is not yet a full smooth 3D metric model.
It does not by itself prove physical electromagnetism or the Standard Model.

Short form:

v39 gives the coefficient ledger.
This file gives the carrier geometry behind that ledger.

1. Minimal geometric object

The v39 carrier is:

$C = Z_{10_port} \times Z_{3_surface} \times Z_{3_trace}$
 $= 10 \times 3 \times 3$
 $= 90 \text{ carrier cells}$

Use coordinates:

(u, s, t)

u in Z10_port	local port coordinate
s in Z3_surface	surface stencil: before / middle / after
t in Z3_trace	trace role: current / retained / activated seam

So the minimal drawn object is a discrete 10 x 3 x 3 carrier block.

The intended interpretation is:

Z10_port:
local width of the first RT-RP port rho=10.

Z3_surface:

minimal scalar surface witness across the port.

Z3_trace:
current / retained / activated seam.

This is the first place where C90 is not merely a number. It is the cell count of the internal readout carrier.

2. Why this is geometry, but not yet full metric 3D physics

This model has:

axes / factors:
Z10, Z3, Z3

cells:
90 discrete carrier positions

roles:
port coordinate, surface stencil, trace state

relations:
current-retained gluing over Q anchors
activated seam layer

It does not yet specify:

Euclidean coordinates
metric distances
continuous embedding
physical lengths
forces
field equations
QED renormalization

Therefore the honest status is:

Discrete carrier geometry: yes.
Full metric 3D physical geometry: not in this file.

This distinction matters. The P0mu_R coefficient uses the discrete readout carrier. It does not require a finished continuum embedding.

3. Port geometry: null, sigma-guard, visible split

Port activation: null -> sigma-guard -> visible primitive split



Figure 1. Local activation sequence for the first RT-RP visible port.

The first local port is constructed around a balanced null:

$\rho = 10 = 2c$, $c=5$

5|5 = RP-null
4|6 = sigma-guard / latent mismatch
3|7 = first visible primitive scalar split

Geometric meaning:

5|5:
no readable surface difference.

4|6:
first imbalance exists, but is still latent/guarded.

3|7:
first primitive visible surface split.

This is the local port geometry behind $\rho=10$.

The direct alternative:

$\rho=8$ gives $4|4 \rightarrow 3|5$ immediately.

That is a direct/external-readout geometry. It lacks the activation step required by RT-RP.

4. Surface layer: C30

For one trace role fixed, the carrier surface is:

$Z_{10_port} \times Z_{3_surface} = 30 \text{ cells}$

This is C30.

It is not a smooth surface. It is a minimal discrete surface stencil over the port.

The three surface positions can be read as:

before / middle / after

or, more abstractly:

left witness / seam witness / right witness

The important point is type, not naming:

C30 is the local port made surface-readable.

5. Trace geometry: current, retained, activated seam

Minimal carrier: $C = Z_{10_port} \times Z_{3_surface} \times Z_{3_trace} = 90 \text{ cells}$

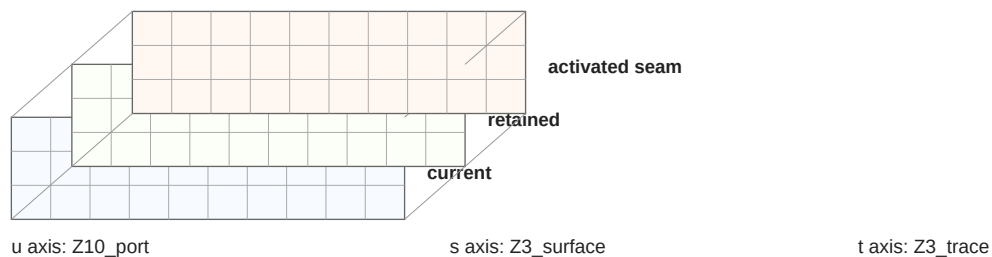


Figure 2. Minimal discrete carrier: ten port positions, three surface witnesses, three trace roles.

RT-RP does not read a single raw process as a full surface. It reads a relation.

The minimal relation carrier has three trace roles:

t = 0: current
t = 1: retained
t = 2: activated seam

Thus:

$C_{90} = C_{30} \times Z_{3_trace}$

This is why $C_{60} = \text{current} + \text{retained}$ is insufficient under RT-RP. It describes an external comparison of two layers, not an internally activated readout.

The activated seam is the third role that makes the relation readable inside RP.

6. Branch-return and boundary book

From the visible split class:

$$\{3, 7\}$$

the first joint branch return is:

$$R = \text{lcm}(3, 7) = 21$$

The first common register between branch return and surface grid is:

$$\begin{aligned} B &= \text{lcm}(R, C30) \\ &= \text{lcm}(21, 30) \\ &= 210 \end{aligned}$$

This is the boundary-address book. It is not a product $21 * 30$. It is the first shared phase/register orbit where both branch-return and surface-grid addresses can be compared.

7. Anchor overlap and G39

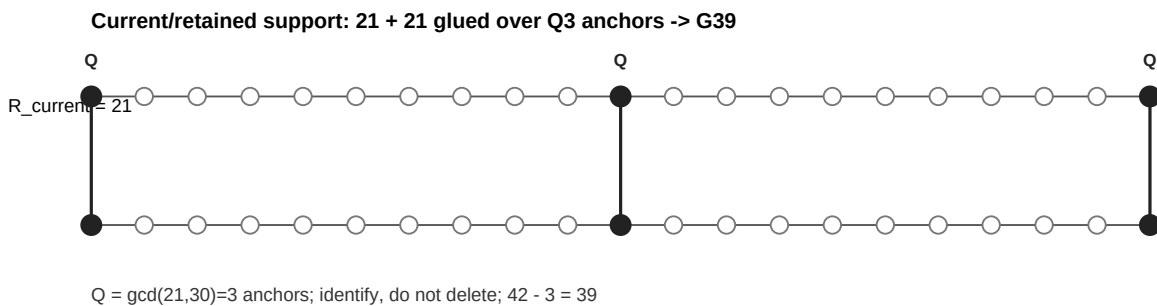


Figure 3. Current and retained R21 supports glued over Q3 anchors, producing G39.

Inside the shared book B210, the overlap of the R21 grid and the C30 grid has size:

$$Q = \text{gcd}(21, 30) = 3$$

These three Q anchors are common boundary anchors.

Current and retained each carry an R21 relation support:

$$\begin{aligned} R_{\text{current}} &= 21 \\ R_{\text{retained}} &= 21 \end{aligned}$$

Before gluing:

$$21 + 21 = 42$$

But the three Q anchors are shared. They must not be counted twice:

$$\begin{aligned} G &= 2R - Q \\ &= 42 - 3 \\ &= 39 \end{aligned}$$

Geometric meaning:

G39 is the current/retained relation support after Q-anchor gluing.

Q is not deleted. Q is not a source-charge triplet. Q is not reused as a later micro-selector. It is the anchor locus that performs the pushout between current and retained support.

8. Seam-core and P0

The P0 seam-core ledger is:

$$\Delta_0 = 2 + 2/\rho - 1/(\rho * G)$$

Geometric reading:

```
2:
  two endpoint incidences of the relation.

2/rho:
  each endpoint must expose through the local port width.

1/(rho*G):
  one rank-one scalar seam-credit over local port support rho and relation support G.
```

So P0 is not just a numerical correction. It is the seam-core accounting of the carrier relation.

9. mu_R as boundary-readable closure

P0 is the latent seam-core. P0mu_R is P0 after the first scalar R-closure is made boundary-readable.

mu_R is idempotent:

```
mu_R(P0)      = P0mu_R
mu_R(P0mu_R)  = P0mu_R
mu_R^2        = mu_R
```

It may only act on the seam-credit:

```
1 -> 1 + epsilon
```

with:

```
active_R = (R-1)/R = 20/21
epsilon  = active_R/B = (20/21)/210 = 2/441
```

Geometric meaning:

```
The marked seam-core uses one R-rank.
The first scalar completion uses the unmarked R-complement.
The closure is addressed over the full boundary book B210.
```

This is the most sensitive geometric refinement. If the correct closure is anchor-local rather than scalar R-complement, the target changes to a different object, such as mu_Q.

10. What the drawing should show

The minimal public drawing should show three layers:

1. Local port:
5|5 null -> 4|6 guard -> 3|7 visible split.
2. Carrier block:
Z10_port x Z3_surface x Z3_trace = 90 cells.
3. Relation support:
current R21 and retained R21 glued over Q3 anchors to form G39.

These three layers are enough to show why the v39 ledger hangs together.

They do not yet show the full RT spiral/tube process, emission process, mass, gravity, or full PP embedding.

11. Relation to the v39 coefficient

The coefficient document uses:

```
lambda_P0mu_R[Theta] = Theta_AB/(2B) * (1 - Delta_mu/C)
```

where:

```
B = 210  from the shared boundary book
C = 90   from the internal carrier block
G = 39   from current/retained gluing over Q3
rho = 10 from the first RT-RP port
```

and:

```
Delta_mu = 2 + 2/rho - (1+epsilon)/(rho*G)
epsilon   = ((R-1)/R)/B
```

Thus the geometry supplies the typed carrier behind the algebra.

12. Failure points exposed by the geometry

The geometric model fails or changes if:

- F1 - RP is external subtraction rather than internal activation.
Then rho=8 and C60 return.
- F2 - the sigma-guard is not required.
Then 3|5 can beat 3|7 as first visible split.
- F3 - current+retained is enough for readout.
Then C90 collapses to C60.
- F4 - Q anchors are deleted or reused.
Then G39 or mu_R changes.
- F5 - mu_R should be anchor-resolved rather than scalar R-complement.
Then P0mu_R is not the right target.
- F6 - the discrete carrier is not the physical EM interface.
Then the coefficient remains internal RT only.

These are real breakpoints. The model should be criticized there, not at the level of arithmetic.

13. Final status

This document gives the carrier geometry behind v39.

It shows why C90, G39, rho=10, B210, Q3, seam-credit,
and mu_R belong to one typed readout structure.

It does not prove the full physical alpha bridge.
It does not prove the Standard Model.
It does not give a full metric 3D PP embedding.

Shortest public line:

The v39 coefficient is a ledger on a discrete RT-RP carrier geometry,
not a free numerological expression.